STAFF REVIEW · H446 002 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q5

Do Now — source answers

1. Integer for the count; Boolean for open/not open.

2. String: the identifier must retain its leading zeros.

3. 128 + 32 + 4 + 2 = 166 .

4. 200 = 128 + 64 + 8 → 11001000 . Eight unsigned bits represent 0–255; 300 needs at least nine bits.

I Do — source worked steps

1. Write +13 in eight bits — CODE:
13 = 8 + 4 + 1
+13 = 00001101 — Both signed representations begin from the positive magnitude.

2. Set the sign bit — CODE:
00001101
→ 10001101 — Keep the seven magnitude bits and change the leftmost sign bit to 1 .

3. Invert, then add 1 — CODE:
00001101
invert → 11110010
add 1 → 11110011

4. Convert the two’s-complement result to denary — CODE:
−128 + 64 + 32 + 16 + 2 + 1 = −13 — Invert-and-add-one negates a value in a fixed width.

We Do — guided CHECK

1. Build the answer
Check: invert 11100110 → 00011001 ; add 1 → 00011010 = 26.

2. Build the answer

3. Build the answer
19: 00010011 → invert 11101100 → add 1 11101101 .

4. Build the answer
Two positive inputs produced a negative stored result.

5. Build the answer

You Do — source mark schemes

1. (a) +37 = 00100101 (1); sign magnitude −37 = 10100101 (1).
(b) Invert +37 → 11011010 (1); add 1 → 11011011 (1).

2. (a) Unsigned: 128 + 16 + 4 + 2 (1) = 150 (1).
(b) Sign magnitude: sign bit 1 with magnitude 16 + 4 + 2 = 22 (1) → −22 (1).
(c) Two’s complement: −128 + 16 + 4 + 2 (1) = −106 (1).

3. CODE:
00101101 (+45)
+ 00011011 (+27)

01001000 (+72)
Correct column addition with carries (1); 01001000 (1); 72 (1); no signed overflow (1). The total is within −128…127.
Starting on the right: column totals are 2, 2, 2, 3, 2, 2, 1, 0 when the incoming carries are included.

4. +29 = 00011101 (1). Invert to 11100010 , then add 1: −29 = 11100011 (1).
CODE:
00010010 (+18)
+ 11100011 (−29)

11110101
Correct addition of +18 and −29 (1); 11110101 (1); −128 + 64 + 32 + 16 + 4 + 1 = −11 (1).

5. (a) 112 and 32 (both for 1).
(b) 10010000 (1).
(c) Correct mathematical total = 144 (1); 144 exceeds the eight-bit signed maximum of +127, so signed overflow occurs (1). The stored pattern represents −112 when interpreted as an 8-bit two’s-complement integer; it is not the correct mathematical sum.
(d) 9 bits (1), with signed range −256…255.
Indicative classroom marking guide; not an official OCR mark scheme. Q1 adapts OCR H446/01 June 2017 Q6(b)(i–ii). Remaining items are original exam-style practice.

Exit Ticket — source answer

1. True. In two’s complement, eight bits represent values from −128 through +127.